Robots Learn to Write via Human–Robot Interaction

Yingli Xu
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology
Wuhan, China
M201872658@hust.edu.cn

Bin Hu
Wuhan National Laboratory for Optoelectronics
Huazhong University of Science and Technology
Wuhan, China
bhu@hust.edu.cn

Jiayuan Yan, Long Chen
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology
Wuhan, China
m201670063@hust.edu.cn, chenclong@yeah.net

Ding-Xue Zhang
Petroleum Engineering College
Yangtze University
Jingzhou, China
zdx7773@163.com

Zhi-Hong Guan
School of Artificial Intelligence and Automation
Huazhong University of Science and Technology
Wuhan, China
zhguan@mail.hust.edu.cn

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***Abstract*—Human–Robot Interaction (HRI) has been extensively investigated in academia, industries, and technology companies. How to teach a robot to write is recognized as a difficult task, involving perception, learning and control of the overall robotic system. This paper develops a robotic handwriting system on a NAO robot by using HRI and Q-learning. In the simulator setup, the human demonstrator first writes to form a desired writing content, with the coordinates of each stroke writing saved; then, by applying the Q-learning algorithm, the virtual NAO automatically learns to find the correct writing order and to optimize the strokes. With the learned knowledge, the real NAO uses its inverse kinematics for generating the joint values from the stroke trajectories, and thus can quickly replicate the letters or words written by the human demonstrator. It is shown that by using the proposed HRI-based handwriting system, the human can naturally and conveniently teach NAO to write many characters and NAO can also have an acceptable writing quality close to the human.**

***Index Terms*—Human–Robot Interaction (HRI), handwriting, NAO robot, Q-learning**

I. Introduction

At present, HRI is getting more and more attention, and still needs further development. In the extensive study of HRI, the ability of the robot for natural, simple and convenient HRI with its users is of central importance [1], [2]. Humanoid robots are increasingly popular in home, industrial and educational institutions because of their natural affinity for being highly similar to human appearance. The robot with an anthropomorphic structures perfectly combines the flexibility of human motion with the strength and precision of cybernetic systems, which allows humanoid robots to perform dexterous human tasks. For example, in the study of humanoid robot assisted education, the robot can be used as a tutor or peer learner, which can effectively improve cognitive and emotional outcomes [3]–[5].

Handwriting is one of the main activities considered in children's language learning activities because it can express spoken words intuitively. Studies [6] have shown that handwriting in the form of automatic letters helps the high-level composition of the text production, although in the primary and secondary school stages, the status of handwriting is relatively low and lacks attention. Therefore, the difficulty of handwriting skills in childhood often has a negative impact on academic performance and self-esteem [7], [8], which still affects approximately 10 to 30% of children. It is worth noting that handwriting, as a complex combination of sports and cognitive skills, is difficult to change once it is formed, so it is important to improve the skills of preschool children from the beginning [6].

In terms of technology, handwriting is one of the most complex tasks performed by the robot, because it requires fine control of servomechanisms. This control involves the dynamic aspects of the writing task, such as speed, force and position [9].

There are a lot of important work in robot-assisted handwriting, but most of them such as [10], [11] are done with devices that are not suitable for natural HRI. Humanoid robots are more appropriate because its appearance is similar to humans, it can improve its integration in the social environment [12]. As one of the most famous HRI humanoid robots, NAO can perform writing tasks assuredly. The work presented in [7], [13] proposes an example of "learning through teaching" to improve children's handwriting skills. While, a system was developed that generates handwritten trajectories for the robot based on the children's presentations, which are expected to represent handwritten Latin alphabets or words with different variability levels. Firstly, the robot increased the variability of the typical erroneous trajectory that represents the child's learning handwriting, while writing the recognized words. After that, children would assess words that written by NAO and present what they think is correct on the tablet. In this way, children can enhance handwriting skill by teaching NAO the correct trajectory. Given the flexibility and scalability of this system, many follow-up studies (for example, [14], [15]) are based on this. In addition, Chandra S et al. ([16]–[18]) developed a robotic autonomous education system that provides one-to-one learning scenarios. The key is to assign

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different levels of learning abilities to robots in order to study the impact of differences in the learning abilities of robots on children's learning.

One problem with the handwriting system proposed by Hood D and Chandra S is that physical writing was not actually performed. In other words, those characters were displayed on the tablet, synchronized with NAO's actions. The biggest difference between trajectory synchronization and physical writing is no pen lifting and pen falling. Therefore, words had to be written in a single stroke, which is usually not consistent with the writing norms of school-age children. A series of factors involved in physical handwriting can affect the dynamics of the writing task. Our research considers the physical handwriting of robots and highly mimics the human writing process to help children form the correct writing specifications. In addition, in order to conform as much as possible to the child's natural state of learning handwriting, and to ensure a certain degree of physical handwriting accuracy, letters or words will be simplified into strokes for writing.

This paper proposes a HRI-based robotic handwriting system. The system is executed with rich voice interaction between NAO and humans. By touching the three sensors on NAO's head, humans will be prompted to choose move forward or backward. NAO physically writes the letters or words that people wrote into the computer. The comparison of human handwriting with NAO shows that the accuracy of NAO's handwriting reproduction is acceptable. We apply the Q-learning algorithm to build a simulator that simulates the real robot's right arm writing scene and finds the correct writing order by continually trying to connect all strokes' coordinates. People can choose to enter that simulator learning stage after handwriting. After a period of learning, you can choose NAO for real training. The contributions of this paper are as follows.

- The adopted NAO robot has complicated kinematics with strong couplings and nonlinearities. We simplify the robot's arm model by reducing the six degrees of freedom into two, thereby reducing the physical handwriting difficulty of the robot.
- Compared to other systems using real robotic handwriting [14], [18], the NAO robot can replicate the human handwriting in a way that is close to the human writing specification, realizing an HRI mode that humans teach robot handwriting.
- Considering the poor glyphs, only the coordinates of starting and ending points of each stroke are saved when the human writing content is saved. With our system, the handwriting reproduction of the robot is better than that written by some human demonstrator. In other words, humans teach robots to write characters, and the robot can in turn identify and correct, which would be important for robot-assisted education.

The remainder of this paper is organized as follows: Section I gives a brief introduction of this research and related work. Section II presents the forward and inverse kinematics analysis for NAO's right hand. Section III use Q-learning algorithm to train handwriting sequence. Section IV realize NAO robot handwriting and learning. Finally, Section V concludes the paper.

II. Robotic Module and Kinematics

A. NAO Robot

NAO is an autonomous humanoid robot developed by Aldebaran Robotics. NAO consists of a number of links, joints and actuators, with 25 degrees of freedom. In this research, the handwriting task mainly involved the right arm, where the arm has 2DOF on the shoulder, 2DOF on the elbow, 1DOF on the wrist, and 1 additional DOF on hand's grasping. Fig. 1 summarizes the link lengths, offsets and rotation angles of the right arm joints that are concerned with the handwriting application. These values are critical for the analysis of forward and reverse kinematics below. In the handwriting task, we only consider two of the six DOF involved in the right arm, that is Right Shoulder Roll(RSR) and Right Elbow Roll(RER). Therefore the Right Shoulder Pitch, Right Elbow Yaw and Right Wrist Yaw angles are kept fixed in the process of handwriting. Eliminating unnecessary joint motion with the motors reduces code complexity, memory usage and power consumption.

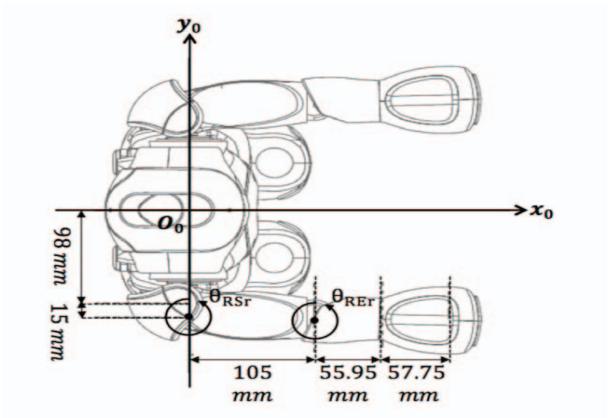


Fig. 1. Top view illustration of the right arm link measurements.

B. Forward kinematics analysis

Robot kinematics focuses on the relationship between the position of the robot end effector and the variable space of the joint, providing a means and method for robot control. Forward kinematics, that is, calculate the position and direction of the end effector based on the joint variables.

Fig. 2 shows link parameter representation of NAO's right arm. The origin O_0 is attached to the coordinate origin at upper left corner of the handwritten white board and computer screen in the same horizontal $x-y$ plane as the right shoulder joint. O_1 is placed at the base of the right shoulder joint. O_2 is attached to the point of actuation of the elbow joint. O_3 is attached to the point of grabbing the pen. Fig. 2 is a simple

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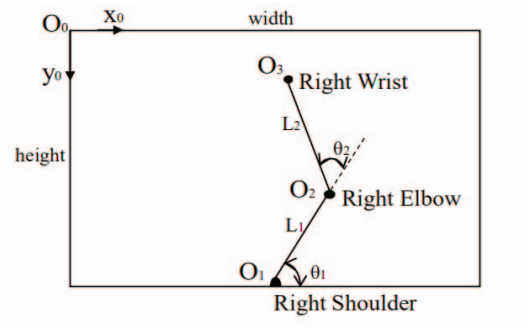


Fig. 2. Brief $x-y$ plane of the NAO robots right arm joints.

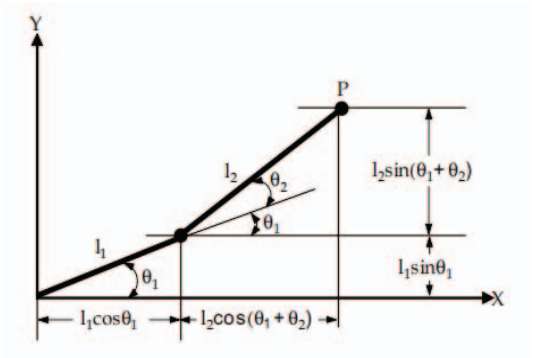


Fig. 3. Solving the inverse kinematics based on trigonometry.

two-dimensional $x-y$ plane, so it should be noted that the $z-$ axis of all joints is vertical $x-y$ plane up.

The link parameter values of Fig. 2 are derived from Fig. 1 as:

- $L_1 = 105 \times 10^{-3} = 0.105m$;
- $L_2 = (55.95 + 57.75) \times 10^{-3} = 0.1137m$;
- $\theta_1 = \theta_{RSr}$;
- $\theta_2 = \theta_{REr}$;
- $\theta_1 \in (-76°, 18°)$;
- $\theta_2 \in (2°, 88.5°)$.

For Fig. 2, the given θ_1 and θ_2 can derive the coordinates of O_3 based on trigonometry:

- $O_1 : x_1 = \frac{width}{2}, y_1 = height$;
- $O_2 : x_2 = x_1 + L_1 \cos\theta_1, y_2 = y_1 - L_1 \sin\theta_1$;
- $O_3 : x_3 = x_2 + L_2 \cos\theta_1 + \theta_2, y_3 = y_2 - L_2 \sin\theta_1 + \theta_2$.

C. Inverse kinematics solutions

The handwritten task performed by the robot is in Cartesian space, and the actuator works in joint space. Cartesian space includes direction matrix and position vector, while joint space is represented by joint angles. Converting the position and direction of the manipulator end effector from Cartesian space to joint space is called inverse kinematics [19]. For Fig. 3, We also analyzed the inverse kinematics based on trigonometry to obtain similar results in as follows:

$$p_x = L_1 \cos\theta_1 + L_2 \cos\theta_1 + \theta_2 \tag{1}$$

$$p_y = L_1 \sin\theta_1 + L_2 \sin\theta_1 + \theta_2 \tag{2}$$

θ_2 can be computed from summation of squaring both (1) and (2). After that, θ_1 can be obtained by deriving the calculated θ_2 into (1) and (2). Results are as follows:

- $\cos\theta_2 = \frac{{p_x}^2 + {p_y}^2 - {L_1}^2 - {L_2}^2}{2L_1L_2}$;
- $\theta_2 = \arccos \frac{{p_x}^2 + {p_y}^2 - {L_1}^2 - {L_2}^2}{2L_1L_2}$;
- $\cos\theta_1 = \frac{p_x(L_1 + L_2\cos\theta_2) + p_y L_2 \sin\theta_2}{{p_x}^2 + {p_y}^2}$;
- $\theta_1 = \arccos \frac{p_x(L_1 + L_2\cos\theta_2) + p_y L_2 \sin\theta_2}{{p_x}^2 + {p_y}^2}$.

III. Q-LEARNING OF HANDWRITING

Reinforcement Learning(RL) is a machine learning paradigm that is especially suitable applying on robots. Based on its own experimental learned knowledge, it is possible to solve problems in a finite time. The basic idea is that an agent(robot) is placed in an environment and can observe the results of its own actions. It can see the environment and concludes its current state s_t. Then the algorithm selects an action a_t. Taking this action can change the environment and the agent's state becomes s_{t+1}. The agent receives a reward r_{t+1} for the action a_t. While processing, the agent keeps trying to maximize this reward. In this work, we use the Q-learning algorithm to help robot learn the handwriting letter.

A. Q-learning algorithm

In the Q-learning paradigm [20], the agent interacts with the environment and performs a set of actions. The environment is then modified and the agent senses new states. In addition, the agent receives an external reward at each time period. In this learning strategy, a goal is defined and the learning process is carried out through trial and error interactions in a dynamic environment. The agent receives rewards or penalties based on actions it takes. s indicate a state and a represents an action. The goal of this learning strategy is teaching agent the optimal control policy, $s \rightarrow a$, to maximize the amount of rewards received over a long period of time. During the learning process, the Q value of each state-action pair $Q(s, a)$, is stored and updated. The Q value represents the usefulness of taking action a being in state s. Q-learning directly approaches to the optimal action-value function, regardless of the current policy. Its update as (3):

$$\begin{aligned} Q(s_t, a_t) \leftarrow\ & Q(s_t, a_t) + \\ & \alpha(r_{t+1} + \gamma max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)) \end{aligned} \tag{3}$$

Where r_{t+1} denotes the reward received at epoch t, α denotes the learning rate and $0 < \gamma < 1$ denotes the discount factor, it is a a measure of how much attention we pay to possible rewards that we might get in the future. Let $Q^\star(s, a)$ denotes the optimal expected Q value of state-action pair (s, a). If each action is executed in every state infinitely, $Q(s, a)$ will converge to $Q^\star(s, a)$.

The Q-Learning algorithm is shown in Algorithm 1.

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Algorithm 1: Q-learning algorithm.

Input: episodes, α, γ
Output: Q

initialize : set $Q(s,a)$ arbitrarily, for each s in S and a in A; set $Q(terminalstate,\cdot)$= 0;
repeat for each episode
  Initialize S;
  repeat for each step of episode
    Choose A from S using policy derived from (e.g. ϵ-greedy);
    Take action A, observe R, S' ;
    $Q(s_t,a_t) \leftarrow Q(s_t,a_t) + \alpha(r_{t+1} + \gamma max_{a'} Q(s_{t+1},a') - Q(s_t,a_t))$;
    $S \leftarrow S'$;
  until S *is terminal state*;
until *all episodes are visited*;

B. Pre-training simulator

We apply the Q-learning algorithm to construct a simulator for robot handwriting pre-training. Then based on this, handwriting training for the humanoid robot.

The simulator is to train the agent(robot) finding the correct order of handwriting. For example, training the agent to write "H" in the correct order. "H" consists of three strokes in the order of writing. We need to save the coordinates of each stroke from starting point to ending point, that is, save the coordinates of three pairs of six points in turn for "H". When writing each stroke, first raise the pen to the starting point of the stroke, then put down the pen, write from starting point to ending point. After reaching the end point, the pen is lifted, the writing of the stroke is completed, the next stroke is also the same step. Before writing training, the simulator environment needs to be set including the upper arm, forearm, joints of the robot and the handwriting area. The coordinates of the saved multiple points are corresponding to the resolution of the computer screen, so we need to convert them to the handwriting area of the simulator. Simulator training writing is finding the correct order to connect all the target points and then get the target letters. The Q-Learning algorithm constructs training simulator. The key factors are:
States: The State set = {current(), achieved[]}:

- current(), current = ((E, S), penState), (E, S) is the angle of robot's Right Elbow Roll(RER) and Right Shoulder Roll(RSR), penState is a string variable: "True" for pen down, "False" for pen up.
- achieved[], achieved = [((E1, S1), penState), ((E2, S2), penState), ...]: store the reward that has been achieved.

Actions: The Action set = { True, False, (E1,S1), (E2,S2)...}.
Rewards: If the n+1th state state ((E, S), penState) is equal to the n+1th item in the list of states constructed according to the points previously saved in the order of writing strokes. At the same time, the nth state is also equal to the nth item in the list of states constructed according to the points previously saved in the order of writing strokes, the bonus value is added to one, and the other cases are subtracted by one.

(E, S) is obtained by inverse kinematics based on the coordinates of each point in the writing area of the simulator. The number of ((E, S), penState) is twice the number of saved points because there are "True" and "False" for penState at each point. The action taken by each state is either changing the penState or the next point. For "H", we needs to save the coordinates of six points, the length of current() is 12. achieved[] keeps expanding until len(achieved)=12, ending the episode. The number of actions that can be taken is 8(6+2).

When the simulator performs handwritten pre-training, several parameters of Q-learning algorithm are set as follows:

α: It is a trade-off measure between the last learned result and this learning result. Setting the alpha too low will cause the robot to care only about the previous knowledge, reducing the accumulation of new rewards. Therefore, we take it 0.5 to balance the previous knowledge with the new reward.

γ: It is a factor that considers future rewards. We take it 0.9 to fully consider the future rewards. In fact, if we reduce it, we will find that the positive reward at the end point cannot be "spread around, which means that the robot may not be able to learn a strategy to reach the end.

ϵ: It is a trade-off measure between exploration and exploitation. We want to ensure that the rewards approximatively continue to grow to the maximum, and once the maximum reward is reached, there will be no sudden individual bad results. Therefore, we use the greedy strategy, which take the ϵ to 0, and it can be accepted from the simulation results.

IV. Experiments

To demonstrate the effectiveness of our approach, we conducted experiments using the NAO robot. In this section, we first discuss the setups of our experiments. Then, we focused on robot handwriting tasks by the comparison between robot handwriting and humans' writing. Finally, the Q-learning algorithm is used to train robots to find correct order of writing to further evaluate the feasibility of applying the Q-learning algorithm to learn handwriting.

A. Experiment Setup

In the writing task, the robot's torso and legs are placed at a constant position to support the robot's sitting posture, and NAO only uses the right arm to perform the task. In order to enhance the perfection of the experiment, we constructed a NAO robot handwriting HRI system, which is shown in Fig. 4.

The entire system is built in the Python 2.7 environment, which contains the Python application programming interface (API) that can call various application modules in naoqi. First of all, NAO enters the initial state and asking human to write into the computer through the mouse. After saving the coordinates of starting and ending points of each stroke by human written, the robot enters the ready state, and the right arm moves to perform the grasping pen operation. Once the preparation is completed, NAO will ask to select the mode to enter by touching one of the three different sensors on NAO's

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head. Touch the front one to enter the robot handwriting stage. Firstly a correction window will pop up to help people put the white board for handwriting to an appropriate place, which is taken with the bottom camera of NAO. Then, the coordinates of the points saved previously in the stroke order are read, and the data are processed in units of strokes by adding state information that the pen needs to be raised or lowered. Since the coordinates of the previously saved points are relative to the resolution of the computer screen, it is necessary to convert these coordinate values into the coordinates of the writing area.

After the conversion is completed, the coordinates are brought into inverse kinematics to solve and obtain the corresponding angle of NAO's RER and RSR. According to the saved joint angle, operate NAO's right arm to write each stroke: lift the arm, move it to the start point, drop the pen, and move it to the end point, so that a stroke is completed. Subsequent strokes repeat these steps until all the them being written. Touch the middle sensor to enter the learning stage. At this stage, firstly let the simulator train 20 episodes observing the learning process and results. Then, touch NAO's head sensor to select the mode again. If the front one is touched, the simulator retrain 5 more. Once the middle one is touched, the robot start train 5 episodes. Each time the rear sensor is touched, the current mode selection ends and returns to the previous mode selection eventually ending the program. See Fig. 4 for details.

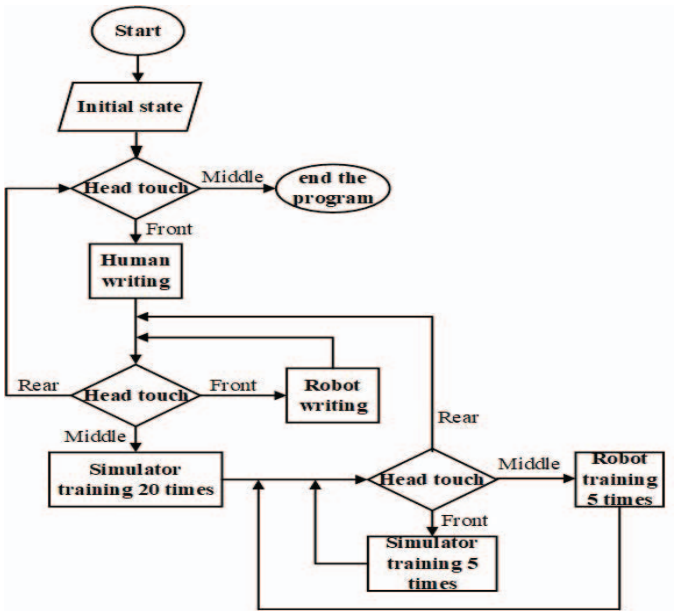
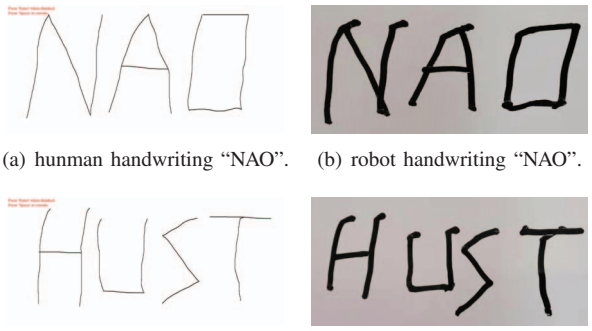


Fig. 4. NAO robot handwriting HRI system.

B. Experimental Results

Next, we experimented with a real NAO robot and got the following results. At the beginning, human write the word "NAO" into the computer through the mouse, then touch the front sensor on NAO's head to enter the robot handwriting stage. Fig. 5 show the process of robot handwritten the word "NAO". We experimented with writing multiple different words into the computer and observing the effect of the robot handwriting and the comparison results between NAO's handwriting and humans' writing are shown in Fig. 6. It can be seen from Fig. 6 that there are some deviations at the head and tail of each stroke of the robot handwriting, which because the joints controlled by the motor have a certain inertia when moving. In general, the effect of robot handwriting is acceptable.



(a) hunman handwriting "NAO". (b) robot handwriting "NAO".

(c) hunman handwriting "HUST". (d) robot handwriting "HUST".

Fig. 6. Human handwriting and robot handwriting contrast

Touch the middle sensor on NAO's head to enter the learning stage, the simulator starts learn the writing sequence of the word "NAO". Fig. 7 and Fig. 8 shows the simulator handwriting training process and result for word "NAO".

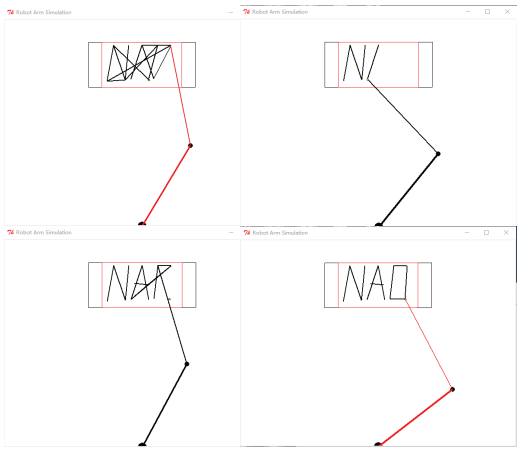


Fig. 7. Simulator handwriting training process for word "NAO".

Fig. 8 shows the change in rewards during the process of training learning the writing sequence of the word "NAO". We can see that the initial reward value is close to the minimum value(-40), which means that the simulator is writing randomly. Then, the reward value gradually increases, and there are few suddenly very bad results in the process. After about 250 sequences, the reward value is increased to a maximum value(40), at which point the simulator has learned the correct writing order, and then the reward value no longer changes and remains at the maximum.

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(a) ready pen. (b) grasp pen. (c) "N". (d) "A". (e) "O". (f) release pen.

Fig. 5. The process of handwriting word "NAO"

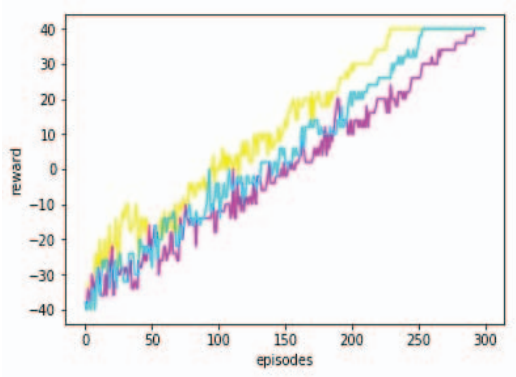


Fig. 8. Rewards of simulator handwriting training for word "NAO".

V. Conclusion

In this paper, we propose an HRI-based robotic handwriting system and test the system on a NAO robot. In this system, a human writes onto a computer firstly, and then NAO replicates the same letters. For illustration, a virtual simulator is constructed of NAO to train and learn a sequence of writing strokes of the letters. After training, the real NAO gets the learning ability and is able to find an optimized writing order. A comparison between the human and robot handwriting trajectories shows that the accuracy of robotic handwriting is close to the human. The virtual simulator helps to find the correct writing order of letters and thus brings less training on the real NAO. These finds reveal that the robot-assisted handwriting can be guaranteed and enhanced via HRI. Future work includes optimizing the performance of the HRI-based robot handwriting system and designing other possible HRI functions on NAO robots.

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